Perceptron

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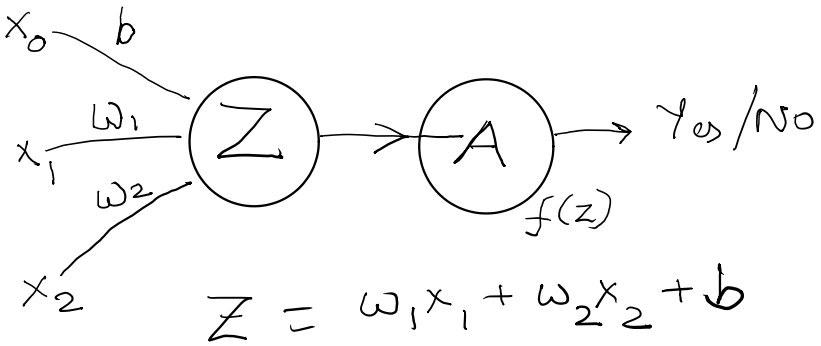
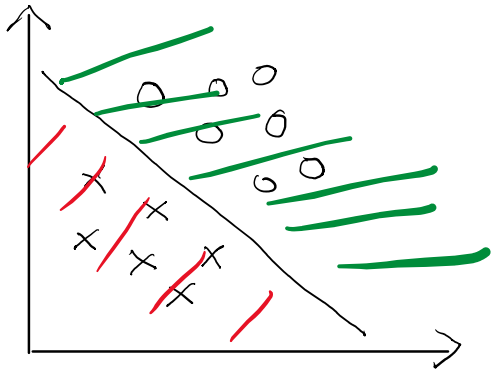
We want to predict if Loan will be approved or not based on 2 features:

- Credit Score (x1)
- Salary (x2)





$$Z = w_1 x_1 + w_2 x_2 + b$$

$$y = f(z) = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{if } z < 0 \end{cases}$$

Loss Function

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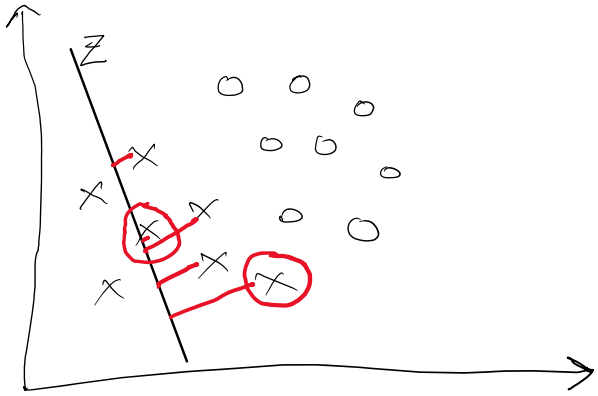

Loss (ex 1) → 5 misclassified data points

$$L(w_1, w_2, b) = \frac{1}{n} \sum_{i=1}^{n} \max(0, -y_i z_i)$$

Only misclassified data points contributes to the loss function

Z = w1 x1 + w2 x2 + b

Wnew = Wold + (LR) * (Loss)

LR -> Learning Rate (ex. 0.01, 0.1)

Perceptron Problem

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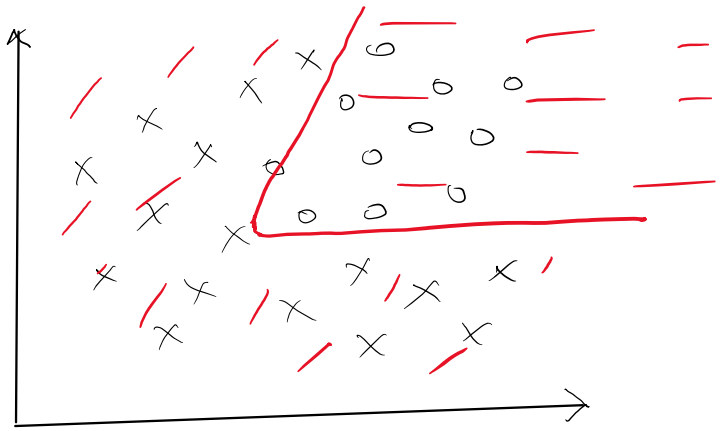
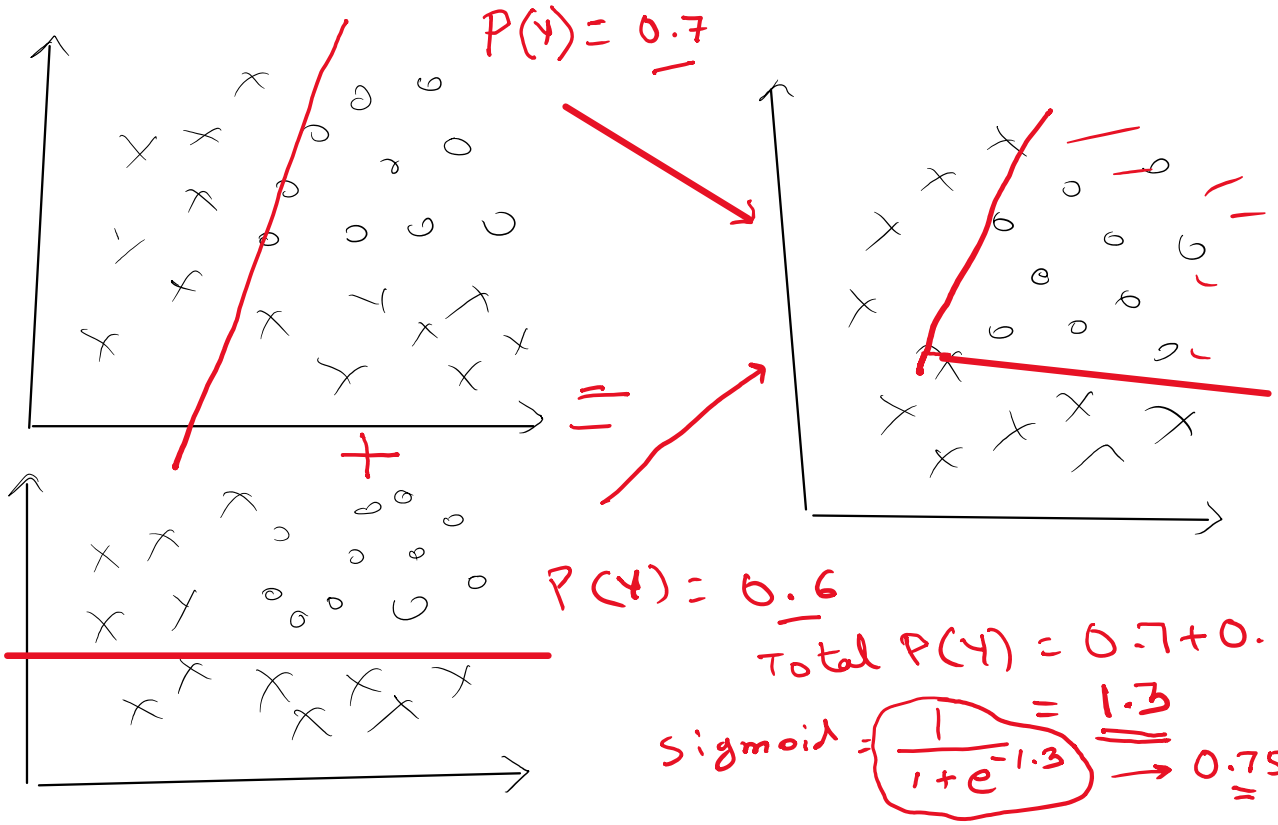
Perceptron can work only where the data points can be separated with a linear model





Multilayer Perceptron

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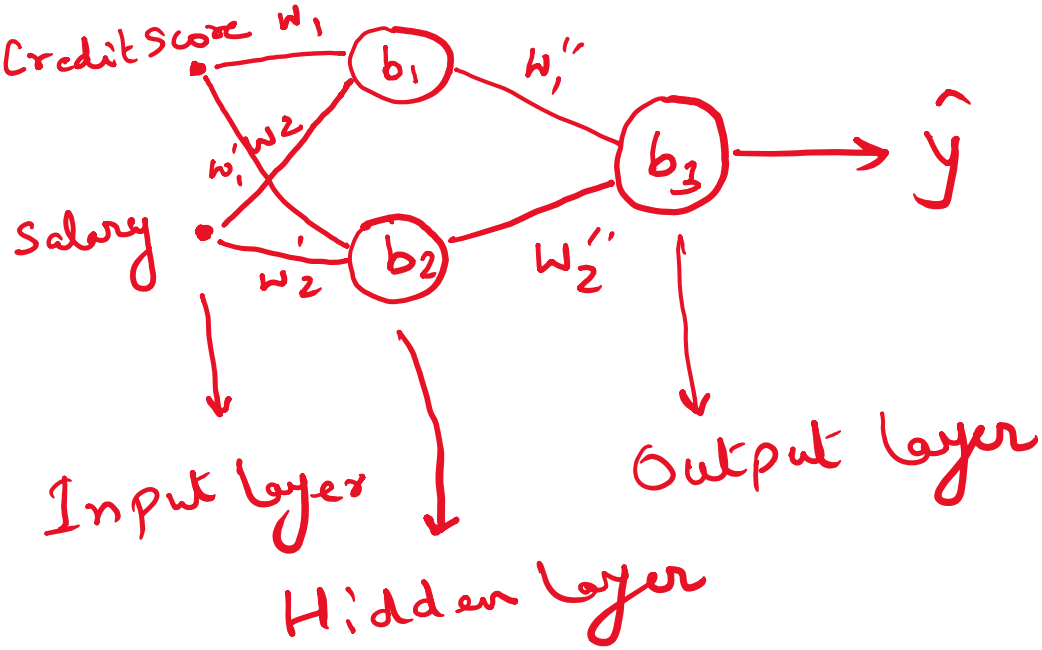

Forward Propagation

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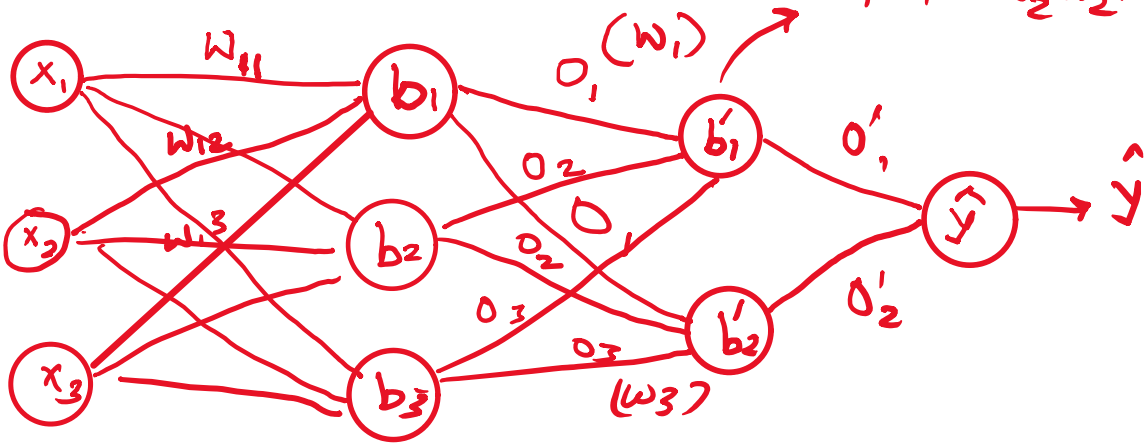
Total → 1000 rows

Credit Score	Salary	N.W	HL
7.8	50K	3M	1
6.1	25K	1M	0

Features:
Credit Score (x1)
Salary (x2)
Net Worth (x3)

Label: Home Loan (Y/N) -> (1/0)

$O_1 w_1 + O_2 w_2 + b_1' = Z'$



After an epoch is completed

→ Loss is calculated

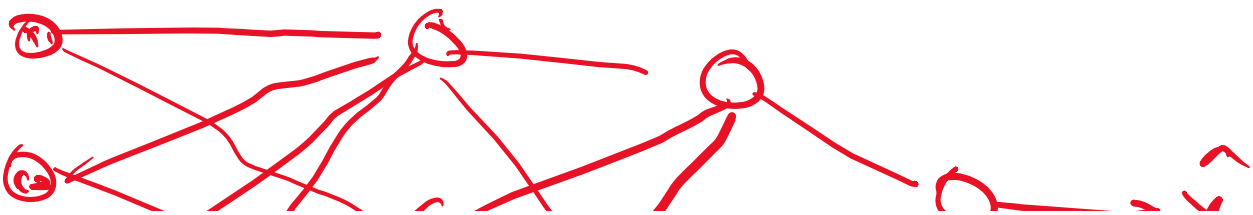
→ Weights & biases are adjusted

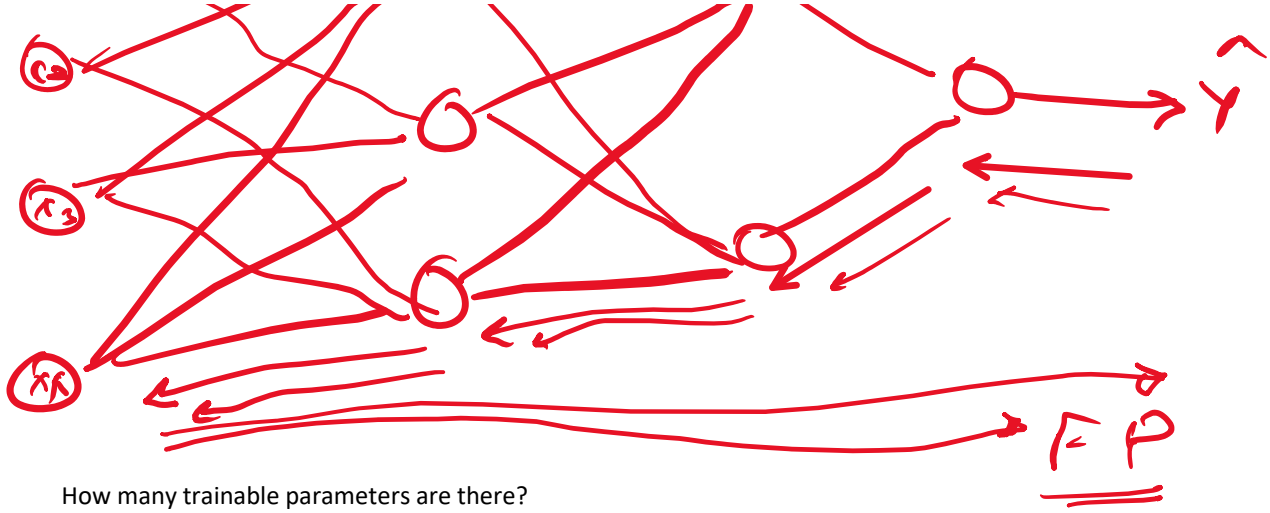
→ Next epoch is run.

How Many trainable parameters are there:

1st Layer -> 3*3 + 3 = 12
2nd Layer -> 3*2 + 2 = 8
3rd Layer -> 2 + 1 = 3

Total = 23





How many trainable parameters are there?

1st Layer = 4 * 3 + 3 = 15
2nd Layer = 3 * 2 + 2 = 8
3rd Layer = 2 + 1 = 3
Total = 26

Forward Propagation -> making predictions

Backpropagation -> Learning for the error, and taking corrective actions.

Backpropagation

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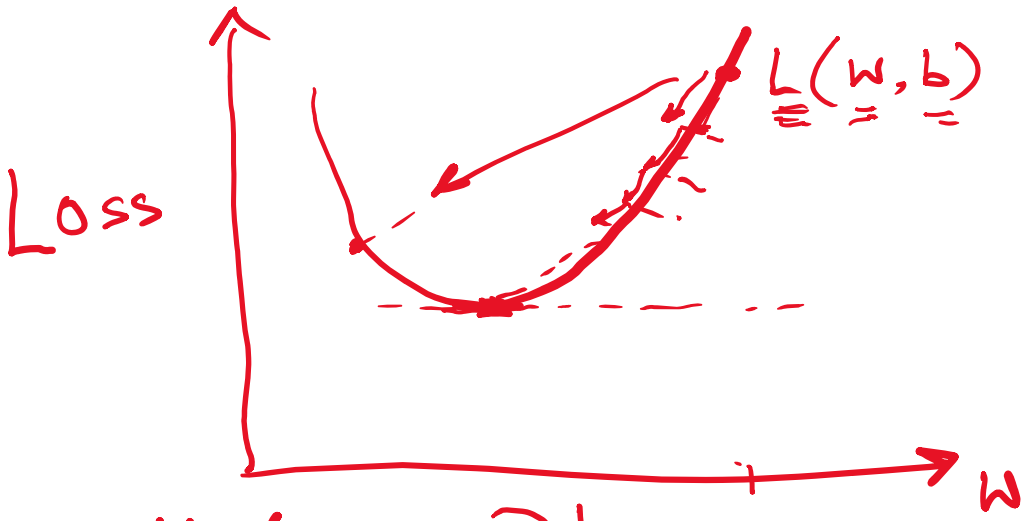
Gradient Descent -> Engine behind the backpropagation

Loss Function -> L (w1, w2, ..., b1, b2, ...)



Gradient $\rightarrow \dfrac{\partial L}{\partial w}$

$w_0 \rightarrow$ initial weight

$$w_1 = w_0 - \eta \times \frac{\partial L}{\partial w}$$

$\eta \rightarrow$ Learning Rate

$b_0 \rightarrow$ initial bias

$$b_1 \rightarrow b_0 - \eta \times \frac{\partial L}{\partial b}$$

Example

loss function: $L = (w - 5)^2$

Loss function: $L = (w-5)^2$

Loss is minimum at $w = 5$

Step 1: Take a random value of w

$$w = 0$$

Step 2: Calculate Gradient

$$\frac{\partial L}{\partial w} = \frac{\partial \left[w^2 + 25 - 10w\right]}{\partial w}$$

$$= 2w + 0 - 10$$

$$= 2w - 10$$

$$= 2(w-5)$$

$$= -10$$

$$w_{new} = w_{old(0)} - \eta \times \frac{\partial L}{\partial w}$$

$$= 0 - 0.1 \times (-10)$$

$$= 1$$

$$w_{old(1)} = 1$$

$$\frac{\partial L}{\partial w} \rightarrow -8$$

$$w_{new(1)} = w_{old(1)} - \eta \times \frac{\partial L}{\partial w}$$

$$w_{new(1)} = w_{old(1)} \quad \eta \frac{\partial}{\partial w}$$

$$= 1 - 0.1 \times (-8)$$

$$= 1.8$$

Slowly it moves towards 5

How does Gradient Descent works with Backpropagation?

For a Linear Model:

Y = mx + b

Y = 3x + 1

Input (x) = 2

Output = 3*2 + 1 = 7

Suppose:
Actual Value = 10
Predicted Value = 7

Loss (Mean Squared Error)

L = [y(actual) - y(pred)]^2

Loss = (10 - 7)^2 = 9 $= (y - y_p)^2 \;/\; (y_p - y)^2$

$(w-3)^2 \rightarrow 2(w-3)$

What Back propagation does?

It computes:

Gradient of loss w.r.t weight
Gradient of loss w.r.t bias

$$y = wx + 1$$

$$\frac{\partial L}{\partial w} \rightarrow \frac{\partial L}{\partial y} \times \frac{\partial y}{\partial w}$$

$$\frac{\partial L}{\partial w} \rightarrow \frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial w}$$

$$= 2 \times (y - y_p) \times x$$

$$= 2(10-7) \times 2$$

$$= 2 \times 3 \times 2$$

$$= 12$$

$$\frac{\partial L}{\partial b} \rightarrow \frac{\partial L}{\partial y} \times \frac{\partial y}{\partial b}$$

$$y = \underbrace{wx}_{0} + \underbrace{b}_{1}$$

$$= (2 \times 3) \times 1$$

$$= 6$$

$$w_{new} = w_{old} + \eta \times \frac{\partial L}{\partial w}$$

$$b_{new} = b_{old} + \eta \times \frac{\partial L}{\partial b}$$

$$w_{new} = 3 + 0.1 \times (12)$$

$$= 3 + 1.2$$

$$= \cancel{1.8}\ 4.2$$

$$b = 1 + 0.1 \times 6$$

$$b = 1 + 0.1 \times 6$$

$$1 + 0.6$$

$$= \cancel{0.4}\ 1.6$$

$$y_{Pred(2)} = \cancel{1.8x + 0.4}\ \ 4.2x + 1.6$$

$$= \cancel{1.8 \times 2 + 0.4} = 4.2 \times 2 + 1.6$$

$$= \cancel{3.6 + 0.4} = 8.4 + 1.6$$

$$= \cancel{4} \qquad = 10$$